

Houssam Zenati

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RESEARCH INTERESTS	I develop statistical theory and algorithms for reliable inference and decision-making with learned models. My work connects nuisance-robust inference, policy learning and adaptive experimentation: controlling the estimation errors that affect a target or decision, and collecting the observations needed to assess it. I am particularly interested in applying these ideas to biology and biomedicine.
PROFESSIONAL EXPERIENCE	Gatsby Computational Neuroscience Unit, University College London, United Kingdom <i>Research Fellow</i> 2025–present Developing methods for semiparametric efficient distributional causal inference, adaptive data collection, and sequential decision-making. INRIA MIND / PreMeDical, France <i>Postdoctoral Researcher</i> 2023–2024 Research on causal inference and mediation analysis, including double machine learning estimators and open-source software development. Criteo AI Lab, France <i>PhD Researcher</i> 2020–2023 <i>Research Engineer</i> 2019–2020 Worked on counterfactual policy learning and online/bandit problems, including optimization methods, contextual bandits, and kernel methods. Institute for Infocomm Research (A*STAR), Singapore 2017–2018 <i>Research Engineer</i> <i>Research Intern</i> Worked on deep representation learning, generative adversarial networks, anomaly detection, and semi-supervised learning, with applications to medical imaging.
EDUCATION	Université Grenoble Alpes / INRIA Thoth, France 2023 Ph.D. in applied mathematics and machine learning. Thesis: <i>Efficient Methods in Counterfactual Policy Learning and Sequential Decision Making</i>. ENS Paris-Saclay / MVA, France 2019 M.Sc. in mathematics, vision, and learning. CentraleSupélec, France 2015–2019 M.Eng. in applied mathematics and computer science.
SELECTED PUBLICATIONS	* marks authors who contributed equally to the paper. Z. Shen*, H. Zenati*, N. Kallus, A. Gretton, K. Khamaru, A. Bibaut (2026). <i>Efficient Inference after Directionally Stable Adaptive Experiments</i>. Preprint. arXiv:2602.21478 H. Zenati, B. Bozkurt, A. Gretton (2026). <i>Kernel Treatment Effects with Adaptively Collected Data</i>. AISTATS. AISTATS 2026 proceedings H. Zenati, A. Gretton (2026). <i>Semiparametric Efficient Test for Interpretable Distributional Treatment Effects</i>. Preprint. arXiv:2605.08034 S. Girard, A. Bibaut, J.-J. Vie, A. Gretton, N. Kallus, H. Zenati (2026). <i>Fast Best-in-Class Regret for Contextual Bandits</i>. UAI. UAI 2026 proceedings H. Zenati, E. Diemert, M. Martin, J. Mairal, P. Gaillard (2023). <i>Sequential Counterfactual Risk Minimization</i>. ICML. arXiv:2302.12120

* marks authors who contributed equally to the paper.

F. El Khoury*, **H. Zenati***, N. Kallus, M. Arbel, A. Bibaut (2026). *Semiparametric Efficient Bilevel Gradient Estimation*. Preprint.

B. Bozkurt, A. Galashov, D. Meunier, Z. Shen, A. Gretton, **H. Zenati** (2026). *Doubly Robust Proxy Causal Learning with Neural Mean Embeddings*. Preprint.

Z. Shen, N. Kallus, D. Meunier, **H. Zenati**, A. Gretton, A. Bibaut (2026). *Nonparametric Instrumental Variable Analysis Without Structural Equations: Debiased Inference on Functionals of Inverse Problems with No Solutions*. Preprint.

A. Bibaut, **H. Zenati**, T. Rahier, N. Kallus (2026). *Functional Natural Policy Gradients*. Preprint.

H. Zenati, B. Bozkurt, A. Gretton (2025). *Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings*. NeurIPS.

B. Bozkurt, **H. Zenati**, D. Meunier, L. Xu, A. Gretton (2025). *Density Ratio-Free Doubly Robust Proxy Causal Learning*. NeurIPS.

H. Zenati, J. Abecassis, J. Josse, B. Thirion (2025). *Double Debiased Machine Learning for Mediation Analysis with Continuous Treatments*. AISTATS.

J. Abecassis, **H. Zenati**, S. Boumaiza, J. Josse, B. Thirion (2025). *Causal Mediation Analysis with One or Multiple Mediators: A Comparative Study*. Psychological Methods.

H. Zenati, A. Bietti, M. Martin, E. Diemert, P. Gaillard, J. Mairal (2025). *Counterfactual Learning of Stochastic Policies with Continuous Actions*. Transactions on Machine Learning Research.

J. Zhou, P. Gaillard, T. Rahier, **H. Zenati**, J. Arbel (2024). *Towards Efficient and Optimal Covariance-Adaptive Algorithms for Combinatorial Semi-Bandits*. NeurIPS.

M. Martin, P. Mertikopoulos, T. Rahier, **H. Zenati** (2022). *Nested Bandits*. ICML.

H. Zenati, A. Bietti, E. Diemert, J. Mairal, M. Martin, P. Gaillard (2022). *Efficient Kernelized UCB for Contextual Bandits*. AISTATS.

H. Zenati, A. Bietti, M. Martin, E. Diemert, J. Mairal (2020). *Optimization Approaches for Counterfactual Risk Minimization with Continuous Actions*. ICLR Workshop on Causal Learning for Decision Making.

P. Mertikopoulos, B. Lecouat, **H. Zenati**, C.-S. Foo, V. Chandrasekhar, G. Piliouras (2019). *Optimistic Mirror Descent in Saddle-Point Problems: Going the Extra (Gradient) Mile*. ICLR.

K. Ouardini, H. Yang, B. Unnikrishnan, M. Romain, C. Garcin, **H. Zenati**, J. P. Campbell, M. F. Chiang, J. Kalpathy-Cramer, V. Chandrasekhar, et al. (2019). *Towards Practical Unsupervised Anomaly Detection on Retinal Images*. MICCAI Workshop on Domain Adaptation and Representation Transfer.

H. Zenati, M. Romain, C.-S. Foo, B. Lecouat, V. Chandrasekhar (2018). *Adversarially Learned Anomaly Detection*. ICDM.

H. Zenati, C.-S. Foo, B. Lecouat, G. Manek, V. R. Chandrasekhar (2018). *Efficient GAN-Based Anomaly Detection*. ICLR Workshop.

B. Lecouat, C.-S. Foo, **H. Zenati**, V. Chandrasekhar (2018). *Semi-Supervised Learning with GANs: Revisiting Manifold Regularization*. ICLR Workshop.

B. Lecouat, C.-S. Foo, **H. Zenati**, V. Chandrasekhar (2018). *Manifold Regularization with GANs for Semi-Supervised Learning*. arXiv preprint.

B. Lecouat, K. Chang, C.-S. Foo, B. Unnikrishnan, J. M. Brown, **H. Zenati**, A. Beers, V. Chandrasekhar, J. Kalpathy-Cramer, P. Krishnaswamy (2018). *Semi-Supervised Deep Learning for Abnormality Classification in Retinal Images*. NeurIPS Workshop on Machine Learning for Health.

TEACHING	Data Visualisation , Université Paris Dauphine – PSL	2021–2022
	Instructor for Master’s students in the IASD program. Covered visualisation as a statistical and algorithmic tool for exploring, summarizing, and communicating complex data, with topics including graphical principles and dimensionality-reduction methods such as PCA, multidimensional scaling, and t-SNE.	
	Advanced Machine Learning , CentraleSupélec	2024–2025
	Teaching assistant alongside Emilie Chouzenoux. Helped students connect the formal ideas in modern machine learning, statistics, and optimization to practical algorithms and empirical behavior.	
STUDENT SUPERVISION	Causal Inference , New York University Paris	2024–2025
	Co-instructor with Judith Abecassis. Taught four sessions, including lectures and recitations, on randomized and observational causal inference, including identification assumptions, weighting, matching, instrumental variables, and regression discontinuity.	
	Teaching philosophy: conceptual clarity, mathematical rigor, and active engagement.	
	<i>Research co-supervision and mentoring at Master’s and doctoral levels.</i>	
	Julien Zhou , Master’s student, with Pierre Gaillard	2023
	Co-supervised research on covariance-adaptive combinatorial semi-bandits, leading to a NeurIPS 2024 publication.	
	Samuel Girard , PhD student, with Aurélien Bibaut	2025–2026
SERVICE	Mentored research on contextual bandits, leading to a UAI 2026 publication.	
	Robin Leman , Master’s student, with Arthur Gretton	2026
	Research supervision and mentoring.	
	Bariscan Bozkurt , PhD student, with Arthur Gretton	2026
	Research mentoring.	
	Sofiane Tanji , PhD student, UCLouvain	2026
	Co-supervising with Yassine Laguel.	
	Workshop organisation	
	Lead Organiser, Advances in Adaptive Experimentation (AAE) Workshop , Gatsby Computational Neuroscience Unit, UCL, 18–19 June 2026. Two-day programme with seven invited speakers, a poster session and an open-problems session. Organising committee: Houssam Zenati, Bariscan Bozkurt, I-Chun Lin and Arthur Gretton.	
	Seminar organisation	
	Organiser of Gatsby’s external machine learning seminars, UCL, September 2025–August 2026.	
	Reading groups	
	Co-organised, with Dimitri Meunier, a reading group on van der Vaart’s <i>Asymptotic Statistics</i> , Gatsby, UCL, 2026.	
	Co-organised, with Fateme Ghayem, a reading group on Wainwright’s <i>High-Dimensional Statistics: A Non-Asymptotic Viewpoint</i> , Inria Paris, 2023–2024.	
	Peer review	
	NeurIPS, ICML, ICLR, AISTATS and TMLR.	

FUNDING AND AWARDS	Netflix workshop grant	2026
	Secured US\$5,000 from Netflix to support the Advances in Adaptive Experimentation (AAE) Workshop at Gatsby, UCL, 18–19 June 2026.	
	IEEE ICDM Student Travel Award	2018
	Support for conference participation.	
	Singapore International Pre-Graduate Award (SIPGA)	2017
	Support for a research attachment at A*STAR, Singapore.	
TALKS	<i>Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings</i> , INRIA Thoth, France	Oct. 2025
	<i>Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings</i> , Symposium on Mathematical Foundations of Trustworthy Learning, Switzerland	Oct. 2025
	<i>Sequential Counterfactual Risk Minimization</i> , ELLIS Unconference, France	Jul. 2023
	<i>Sequential Counterfactual Risk Minimization</i> , INRIA Lille, France	Jun. 2023
	<i>Nested Bandits</i> , ICML, United States	Jul. 2022
	<i>Efficient Kernelized Contextual Bandits</i> , virtual talk	May 2022
	<i>Optimization Approaches for Counterfactual Risk Minimization with Continuous Actions</i> , virtual talk	Apr. 2020
	<i>Adversarially Learned Anomaly Detection</i> , ICDM, Singapore	Nov. 2018
	<i>Efficient GAN-Based Anomaly Detection</i> , DL2.0 Workshop, I2R, A*STAR, Singapore	Apr. 2018
SOFTWARE	med_bench : Python package for causal mediation analysis with a unified implementation of common estimators and flexible nuisance models.	
	Open-source research code:	
	• adaptive-KTE • counterfactual-policy-mean-embedding • double-debiased-machine-learning-mediation-continuous-treatments • sequential-counterfactual-risk-minimization • Efficient-Kernel-UCB • Nested-Exponential-Weights • optimization-continuous-action-crm • Adversarially-Learned-Anomaly-Detection • Efficient-GAN-Anomaly-Detection	